

This question paper contains 4 printed pages]

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S.No. of Question Paper : 1196

Unique Paper Code : 2343010010

Name of the Paper : Data Privacy

Name of the Course : B.Sc. (Hons.) Computer Science

Type of the Paper : DSE NEP-UGCF

Semester : V

Duration : 3 Hours

Maximum Marks : 90

(Write your Roll No. on the top immediately on receipt of this question paper.)

The question paper consists of two sections.

Section A is compulsory.

Attempt any *four* questions from Section B.

All parts of a question must be attempted together.

Section A

1. (a) What ethical principles should organizations follow when collecting user data ? 3
- (b) What is a masquerade attack ? Is it an active or a passive attack ? Justify your answer. 3
- (c) Explain the following terms with respect to a hash function : 3
 - (i) Preimage resistant
 - (ii) Pseudorandomness.

P.T.O.

- (d) Which of the following could be considered personally identifiable information (PII) ?
Justify your answer : 3
- (i) Date of birth
 - (ii) Device ID
 - (iii) Publicly available Job Title.
- (e) Explain the concept of the 'Right to be Forgotten'. What are the conditions under which an individual can request the erasure of their personal data ? 3
- (f) Discuss the following terms with respect to GDPR : 3
- (i) Natural Person
 - (ii) Legal Person
 - (iii) Data Subject.
- (g) Why is it difficult to achieve a perfect balance between data utility and privacy in anonymization processes ? 3
- (h) What is the need of anonymizing data ? 3
- (i) Can digital signatures be forged ? If not, why ? 3
- (j) What is a security service ? Explain any two security services. 3

Section B

2. (a) Explain the CIA triad with respect to Information Security. Additionally, discuss how the concepts of authenticity and accountability complement these objectives. 7
- (b) What is the difference between a threat and an attack ? Compare and contrast active and passive attacks. 8

3. (a) (i) Classify the following attributes into Explicit Identifiers (EI), Quasi-Identifiers (QI), Sensitive Data (SD), or Non-Sensitive Data (NSD) : 7
- Email ID
 - Income Level
 - Gender
 - Voter ID
- (ii) Discuss why it is crucial to anonymize QI and SD before sharing data publicly.
- (iii) Provide two methods that could be used to anonymize Quasi-Identifiers in this dataset.
- (b) Explain cryptographic hash function. Also, discuss the purpose of using a cryptographic hash function in digital signatures and message integrity ? 8
4. (a) Describe the potential harms caused by loss of privacy. What is the need for separating consumer protection issues from data protection issues ? 7
- (b) A healthcare company developed a machine learning model to predict patient risk factors using data from hospitals. However, the dataset was skewed, with more data from wealthier, majority-group patients, leading to less accurate predictions for minority and low-income groups. 8
- (i) Was the data collection process ethical ? Why or why not ?
- (ii) How can bias in the data lead to discrimination in the model's predictions ?
- (iii) What steps can the company take to reduce bias and improve fairness in the model ?

5. (a) Elaborate and discuss DPDPA. Define a personal data breach and explain the general obligations of a Data Fiduciary to prevent and report such breaches. 7
- (b) Discuss Model Surveillance System. Explain its five steps with the help of a neat diagram. 8
6. (a) How is anonymization done in Multidimensional Data ? What are the challenges associated with privacy preservation of multidimensional data ? 7
- (b) Define Data Privacy. Also, explain the role of all the stakeholders of data privacy in an organization with the help of an example. 8